

USER MANUAL

EnergyDashboard

An Ontario electricity-market observability platform — real-time prices, 5CP peak forecasting, Global Adjustment exposure, and private bill analysis, in one place.

Industrial & Commercial

Retail & Homeowner

• Live IESO data

Privacy-first

Version 1.0 · Six tabs, two audience sections · Not affiliated with the IESO

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1 · Getting started

EnergyDashboard is a web app — nothing to install. Open it in any modern browser on desktop or phone.

The two sections

The header groups six tabs into two audiences. Pick the row that matches you:

Industrial & Commercial

Overview · Nodal · Peak Forecast · GA Exposure. For Class A (ICI) participants, energy managers, and market operators.

Retail & Homeowner

Conservation · Usage Review. For households, renters, and small-business / building managers.

Switching tabs keeps your work

Anything you enter — an uploaded meter file, bill photos, comparator inputs — stays put when you move between tabs and come back. Nothing is lost until you reload the page.

Light / dark theme

Use the sun / moon button at the top-right to switch themes. Dark is the default; your choice is remembered on this device.

Live vs mock data

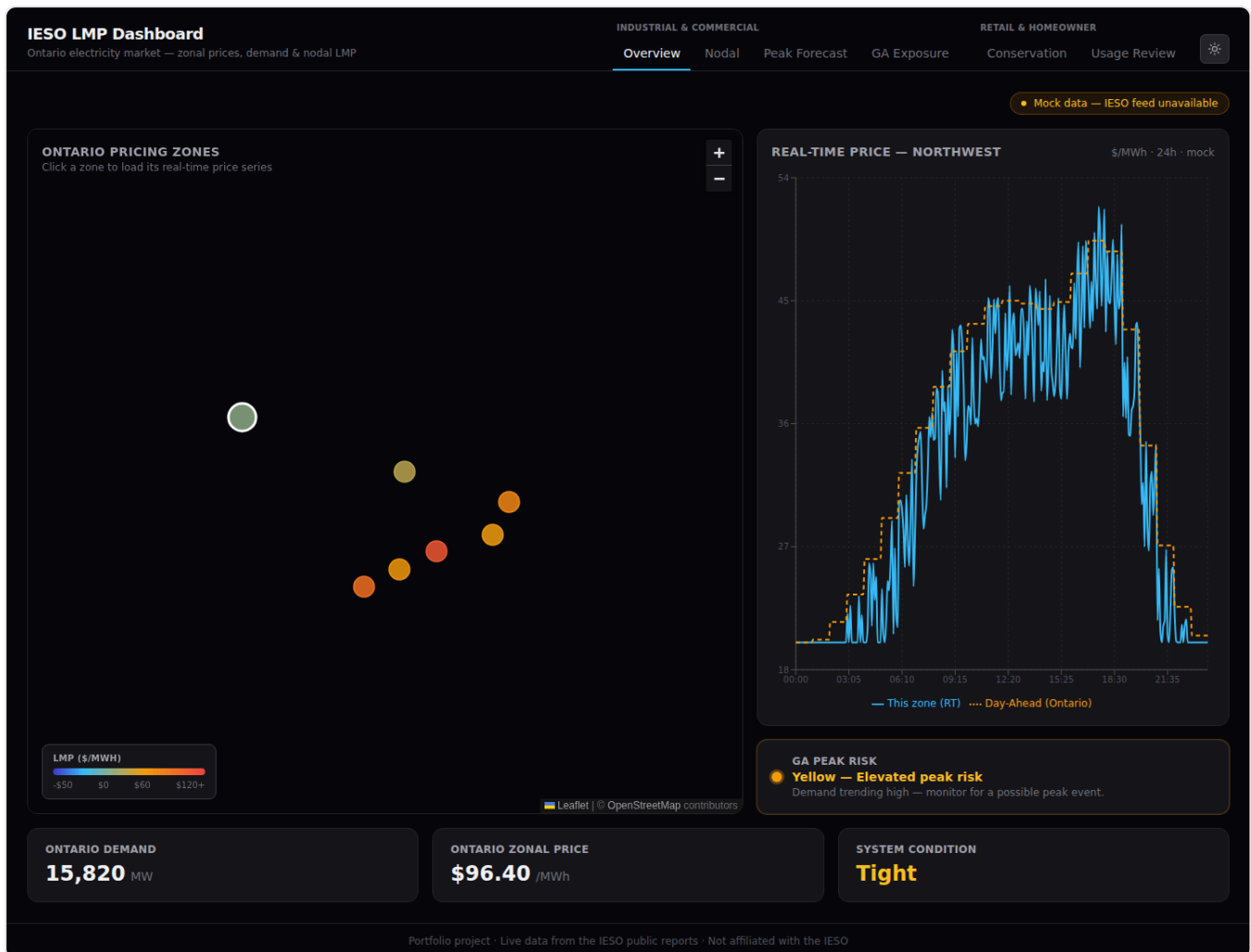
The Overview and Nodal tabs read live IESO market reports. A coloured pill tells you which feed you're seeing:

BADGE	MEANING
• Live (green)	Data is coming straight from the IESO public reports.
• Mock (amber)	The live feed was unreachable, so realistic sample data is shown so the UI still works. Try again shortly.

Tip. Real-time market reports update roughly every five minutes. The app re-fetches on that cadence automatically — no need to refresh the page.

2 · Overview Industrial & Commercial

The "what's happening right now" glance — the whole Ontario market made legible in five seconds.



The Overview tab: zone map, 24h price curve, GA peak-risk, and the demand / price / condition bar.

What you're looking at

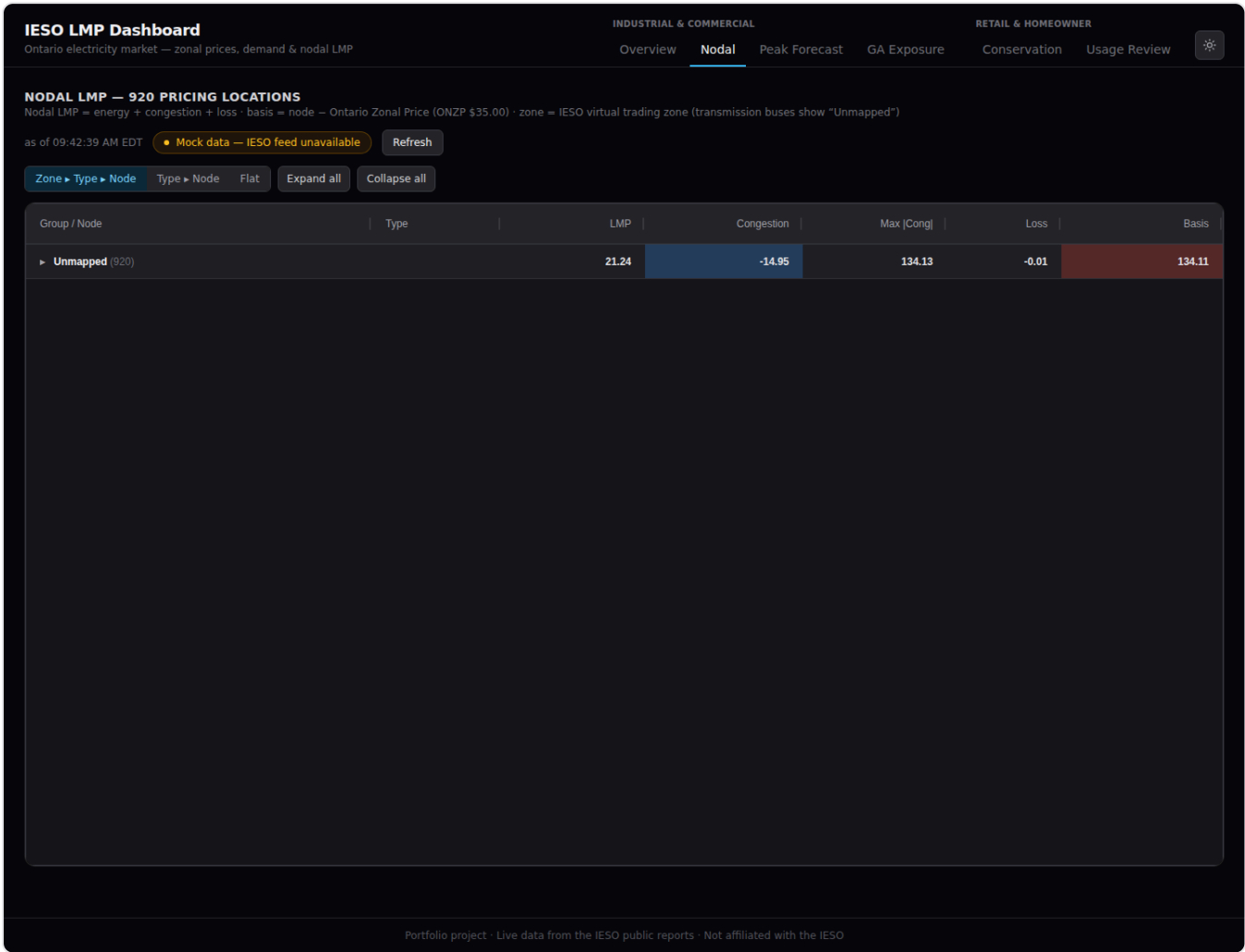
- **Ontario pricing-zone map** — the 7 plotted zones, each dot coloured by its Ontario Zonal Price on a blue → amber → red scale.
- **Real-time vs Day-Ahead chart** — a rolling ~24h of the selected zone's real-time price against the Ontario day-ahead price.
- **GA Peak Risk** — a green/yellow/red indicator of how likely the current hours are to become a coincident peak.
- **Stat bar** — Ontario demand (MW), the headline zonal price, and the system condition (Normal / Tight).

How to use it

1. Click any zone dot on the map to load that zone's real-time price series in the chart.
2. Watch the **GA Peak Risk** tile — when it turns yellow or red, jump to **Peak Forecast** for the detail.

3 · Nodal Industrial & Commercial

The depth view — every pricing node on the grid, decomposed into its parts.



The Nodal grid: 900+ pricing locations pivoted Zone → Type → Node.

What you're looking at

Since Market Renewal (May 2025), Ontario prices energy at hundreds of individual nodes. Each node's Locational Marginal Price breaks down as **LMP = energy + congestion + loss**, and its **basis** is how far it sits from the province-wide Ontario Zonal Price (**basis = LMP – OZP**).

How to use it

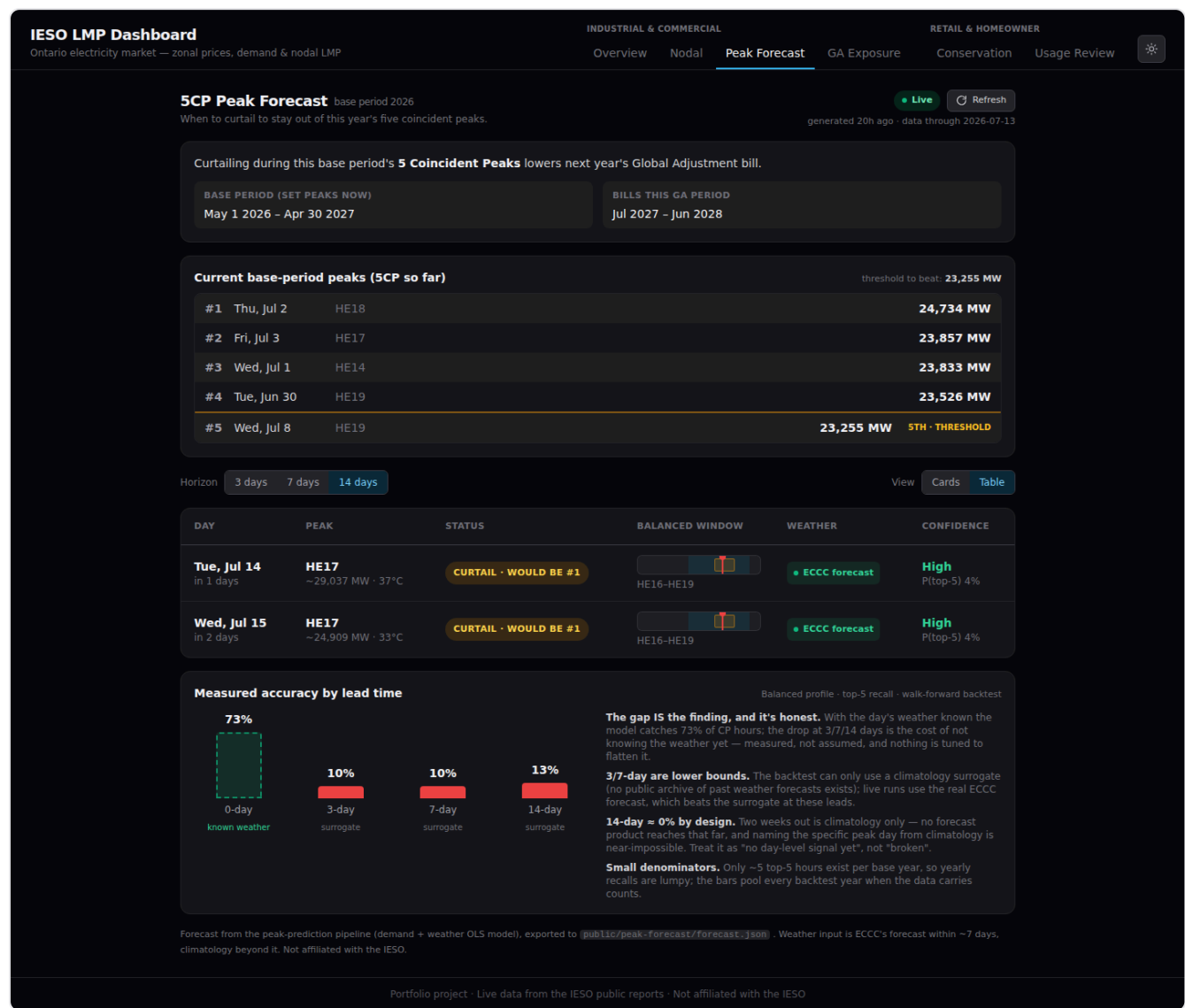
- 1. Use the grouping buttons (**Zone ► Type ► Node**, **Type ► Node**, or **Flat**) to pivot the table.
- 2. **Expand all** / **Collapse all** to open or close every group at once.
- 3. Read the **Congestion** and **Basis** columns to see *where* on the grid the price pressure is.

Note. Nodes the map can't place in a virtual zone appear under "Unmapped" — coverage of the published node-to-zone map is roughly 93%.

4 · Peak Forecast

Industrial & Commercial

The flagship signal — which upcoming hours are likely to become one of this base period's five coincident peaks, and whether they're worth curtailing.



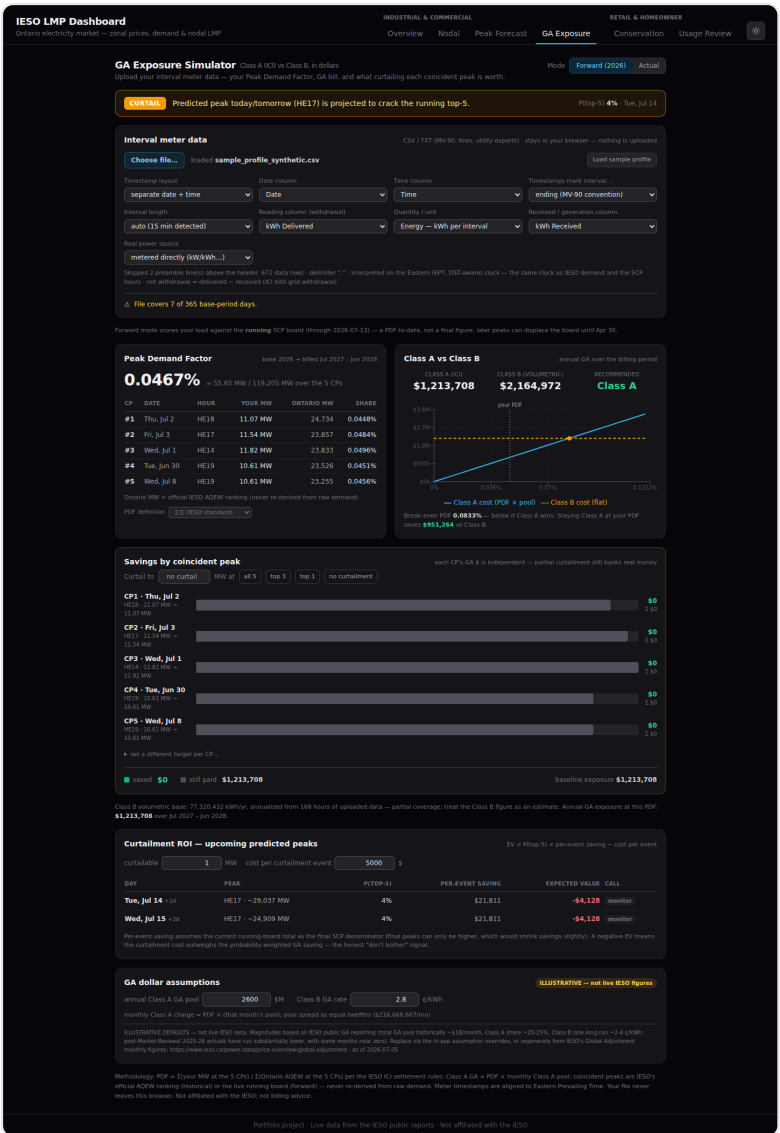
- **Measured accuracy by lead** — honest, backtested recall at each lead time.

Read this honestly. 14-day accuracy sits near zero *by design*: two weeks out there is no real weather forecast, only climatology, so the model can't name the exact peak day. The panel shows the known-weather ceiling next to it so you always see the real ceiling, not a flattering number.

Live vs Sample. A green **Live** pill means real pipeline output. An amber **Sample data** pill means the shipped example is showing (before the first real run of your deployment).

5 · GA Exposure Industrial & Commercial

The money tab — turn your own interval meter data into dollars: your Peak Demand Factor, Class A vs Class B exposure, and what curtailing each peak is worth.



Peak Demand Factor, Class A vs Class B with break-even, per-peak savings, and curtailment ROI.

Your data never leaves your browser. The meter file you upload is parsed and analysed entirely on your device. Nothing is uploaded, stored, or logged. See §8.

How to use it

1. Click **Choose file...** and select your interval meter export (MV-90, Itron, or a utility CSV). No file handy? Click **Load sample profile** to explore with realistic data.
2. Confirm the column mapping (date, time, reading column, interval length, units). The tab auto-detects most exports and tells you what it inferred.
3. Read your results:
 - **Peak Demand Factor** — your MW share across the five coincident peaks.

- **Class A vs Class B** — annual GA dollars each way, with the **break-even PDF** that flips which class wins.
 - **Savings by coincident peak** — set a curtailment target and see the banked saving per CP.
 - **Curtailment ROI** — expected value of curtailing each upcoming predicted peak, using its P(top-5).
4. Switch **Forward (this base period)** vs **Actual** to score against the live running board or a finished period's official peaks.

A negative ROI is a real answer. When the probability-weighted saving doesn't beat the cost of a curtailment event, the ROI shows negative and the call is **monitor** — an honest "don't bother" rather than a nudge to act.

GA pool and rate assumptions are illustrative and adjustable in the "GA dollar assumptions" panel; the settlement math follows the published ICI rules. Not billing advice.

6 · Conservation Retail & Homeowner

Ontario's conservation and billing programs, organised by what you're trying to do — plus a rate-plan comparator that tells you which plan is cheapest for how you actually use power.

IESO LMP Dashboard
Ontario electricity market — retail prices, demand & nodal LMP

INDUSTRIAL & COMMERCIAL
Overview Nodal Peak Forecast GA Exposure Conservation Usage Review

RETAIL & HOMEOWNER
Conservation Usage Review

Conservation & Billing Navigator rebates · rate plans · demand response
Ontario conservation and billing programs, organized by what you're trying to do. updated yesterday

All Residential Business

Rate plan comparator

Monthly kWh: 1000
Off peak %: 65
Mid peak %: 18
On peak %: 17

State of usage you could move overnight (the ILD):

Time-of-Use ☒ Ultra-Low Overnight ☐ Tiered

Illustrative rates — verify at OEB

Time-of-Use	\$75/mo
Ultra-Low Overnight	\$43/mo
Tiered	\$76/mo

Cheapest for this profile: **Time-of-Use** — about **\$51/yr** versus the priciest plan for you. Energy charges only (ILD overnight: 3% vs. on peak 39.1¢), after the 24% DER credit, delivery/regulatory charges excluded. Illustrative — confirm current rates with the OEB.

BILLING & RATE PLANS

Rate Plan Optimization (YOU vs. ULO vs. Tiered) Residential
HOMEOWNERS TIME-SHIFTING LOAD (EV CHARGING, POOL PUMPS, LAUNDRY)
Ultra-Low Overnight (ULD) prices overnight energy at roughly 3.2¢/kWh versus about 39.1¢/kWh on-peak — a large gap that rewards moving flexible load to nights.
ev · time-of-use · ulo · tiered · billing
[Compare my rate plans](#) [verify at source](#)

Ontario Electricity Rebate (OER) Residential Business
GENERAL BILL RELIEF — REDUCING YOUR BILL ACCURATELY
An automatic ~23.5% pre-tax credit applied to eligible residential and small-business bills. No application needed. It appears as a line item.
rebate · bill-relief · oer · credit
[See it in the comparator](#) [verify at source](#)

REBATES & UPGRADES

Home Renovation Savings Program (HRSP) Residential
HOMEOWNERS PLANNING INSULATION, HEAT PUMPS, OR WINDOW UPGRADES
Two paths: a 'Single Upgrade' path (no energy audit required) and a 'Bundled' path that requires a pre- and post-work EnerGuide audit but unlocks larger, stacked rebates.
heat-pump · insulation · windows · rebates · energuide
[Open program page](#) [verify at source](#)

Energy Affordability Program (EAP) Residential
INCOME-ELIGIBLE HOUSEHOLDS NEEDING EFFICIENCY UPGRADES
Up to 100% cost coverage for measures like attic insulation, draft-proofing, and efficient appliances for qualifying households.
income-eligible · insulation · appliances · affordability · free
[Check eligibility](#) [verify at source](#)

Save on Energy Retrofit Program Business
FACILITY MANAGERS UPGRADE HVAC, LIGHTING, OR DERs
Prescriptive incentives for HVAC, lighting controls, and PTHPs. The 2026 update split PTHP tiers (new construction vs. replacement) and adjusted small/medium solar DER incentives.
hvac · lighting · der · retrofit · prescriptive
[Open program page](#) [verify at source](#)

DEMAND RESPONSE

Peak Perks Residential
HOUSEHOLDS WITH CENTRAL A/C AND A SMART THERMOSTAT
About \$75 to enroll and roughly \$20/year to allow small summer peak thermostat adjustments (up to ~2°C). You can override any event.
thermostat · summer · demand-response · ac · peakcover
[Open program page](#) [verify at source](#)

Small Business Peak Perks Business
DEMAND RESPONSE FOR SMALL COMMERCIAL SITES
Open to businesses with roughly 50 or fewer on-site employees; mirrors the residential smart-thermostat program with summer peak adjustments.
thermostat · demand-response · small-business · summer
[Open program page](#) [verify at source](#)

TRACKING & DATA

Global Adjustment (GA) Tracking Business
PP&A SETTLEMENTS, AND OPERATIONAL FORECASTING
Not a subsidy — but tracking the monthly Class B GA rate alongside conservation activity is the single highest-value data point for this audience. For Class A exposure, use the GA Exposure tab.
ga · settlements · forecasting · class-b · tracking
[Open GA Exposure tab](#) [verify at source](#)

Credited from OEB, IESO Save on Energy, and EnerGuide Star program pages — organized by use case, not mirrored verbatim. Details marked "verify at source" are 2026 reference points and can change; always confirm eligibility and amounts on the official page before acting. Refreshed weekly. Not aligned with the IESO, OEB, or any utility.

Portfolio project · Live data from the IESO public reports · Not affiliated with the IESO

Rate-plan comparator on top; the program catalog grouped by use case below.

Finding a program

1. Filter by audience — **All**, **Residential**, or **Business**.
2. Search by keyword (e.g. *heat pump*, *thermostat*, *EV*) to narrow the catalog.
3. Each card shows the key detail, tags, and a link to the official program page. Details marked *verify at source* are 2026 reference points — always confirm eligibility and amounts on the official page before acting.

Comparing rate plans

1. Enter your **monthly kWh** and how it splits across off / mid / on-peak.

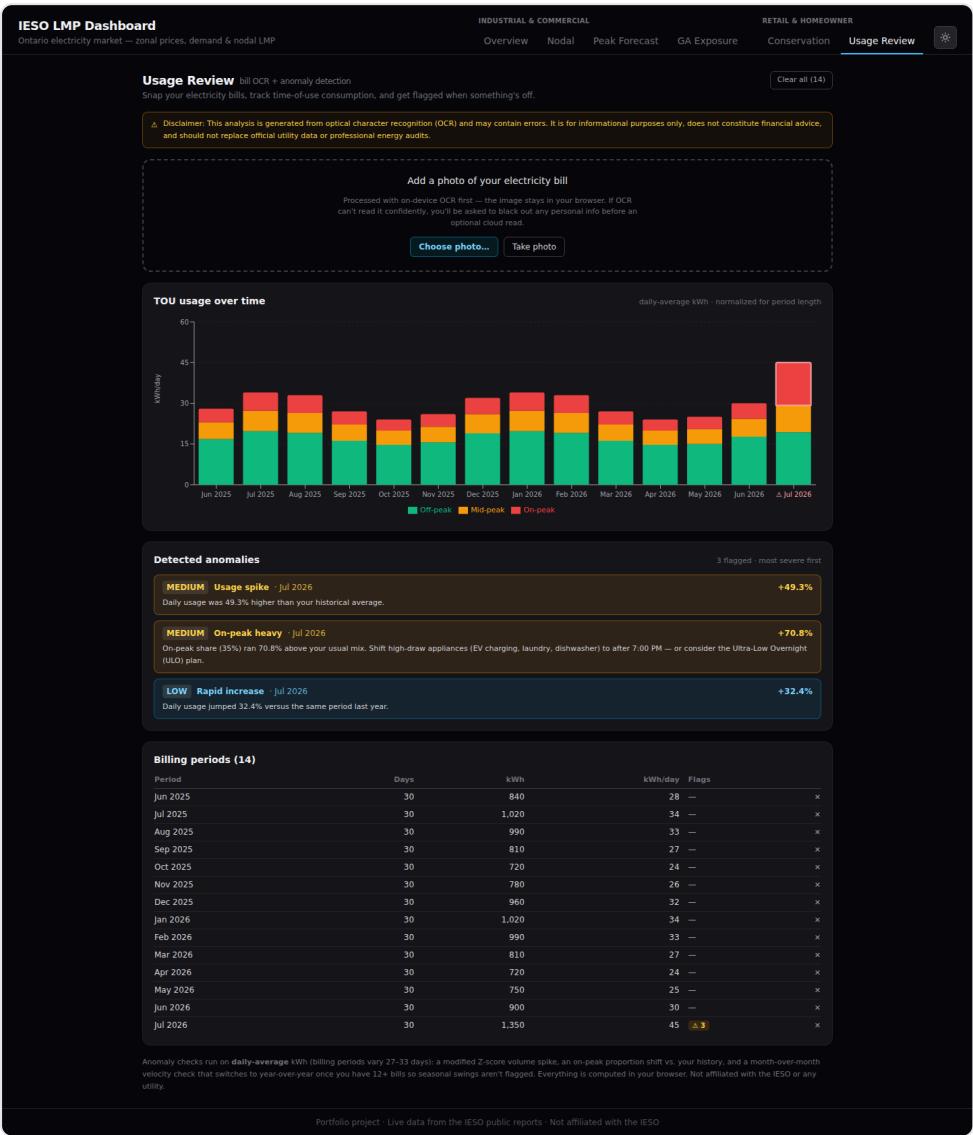
2. Drag the **overnight-shift slider** to model how much load you could move to overnight (this is what makes Ultra-Low Overnight pay off).
3. The bars show each plan's monthly cost after the Ontario Electricity Rebate; the cheapest is marked ✓, with the annual gap versus the priciest plan.

Illustrative rates. The plan structure (TOU / ULO / Tiered windows) is correct; the cents-per-kWh are reference points badged *illustrative* until a live OEB feed is wired. Confirm current prices with the OEB.

7 · Usage Review

Retail & Homeowner

Snap a photo of your electricity bill and get it read, charted, and checked for anomalies — entirely in your browser.



Time-of-use usage over time, detected anomalies explained in plain language, and the billing-period table.

Your bill stays on your device. Reading happens with on-device OCR — the image never leaves your browser. Only if OCR can't read a bill confidently are you offered an optional cloud read, and only *after* you black out any personal details yourself. See §8.

How to use it

1. Click **Choose photo...** or **Take photo** (on a phone). No bill handy? Click **Load sample bills** to see it work.
2. The app reads the meter ID, billing dates, and off/mid/on-peak kWh, then asks you to confirm or correct them.
3. Add more bills over time — they group by meter and build a timeline.

What it flags

ANOMALY	WHAT IT MEANS
Usage spike	Your daily usage this period is well above your own historical average.
On-peak heavy	You're using more of your power during expensive on-peak hours than usual — with a tip to shift load after 7 PM or consider ULO.
Rapid increase	A sharp month-over-month jump — or, once you have a year of bills, a year-over-year jump so seasonal swings don't false-trigger.

Analysis is generated from OCR and may contain errors. It's informational only, not financial advice, and doesn't replace official utility data.

8 · Privacy & your data

The two most sensitive things you can bring — your interval meter data and a photo of your bill — never leave your browser by default. This is built into how the app works, not just promised.

GA Exposure

Your meter CSV is parsed and analysed entirely on your device with pure client-side code. There is no upload, no storage, and no analytics tracking of your data.

Usage Review

Bills are read with on-device OCR. The image stays in your browser and is held only in memory for the current session.

The one optional exception — and how it's contained

If on-device OCR can't read a bill confidently, you may *choose* a cloud-assisted read. Before anything is sent, you black out personal details on a redaction canvas. The server route that performs the read is **ephemeral by design**: it parses the image in memory, returns the extracted numbers, and terminates. It writes nothing to any database, file, or log, and it never records anything derived from your image or personal information. This path is entirely optional — the on-device reader works without it.

In short. No accounts. No server-side store of your energy data. No third-party analytics on what you upload. Reload the page and your session data is simply gone.

9 · Troubleshooting & FAQ

QUESTION	ANSWER
The badge says "Mock data."	The live IESO feed was briefly unreachable; the app shows realistic sample data so it still works. It re-tries automatically — check back in a few minutes.
My meter file didn't map correctly.	Use the column-mapping dropdowns in the GA Exposure tab to point at the right date, time, reading, and unit columns. The panel tells you what it inferred and how many rows it read.
OCR misread my bill.	Confirm-and-correct the fields before saving — you're always shown the extracted values first. For a tough photo, use the optional redact-then-cloud-read fallback.
Are the rate figures current?	The plan structure is correct; the cents-per-kWh are illustrative reference points (badged as such). Confirm live prices with the OEB.
Why is 14-day forecast accuracy so low?	By design — there's no real weather forecast two weeks out, only climatology. The app shows this honestly rather than inflating the number.
Does this bill me or connect to my utility account?	No. It's an observability tool. It reads public market data and whatever <i>you</i> choose to bring; it is not connected to any billing system.

EnergyDashboard User Manual v1.0 · Ontario electricity-market observability · Portfolio project, not affiliated with the IESO, OEB, or any utility
· All market data is public. Figures shown are illustrative unless a live feed is indicated.